PART 1 | Basic principles

CHAPTER 4

Tissue engineering and regenerative therapies

Learning objectives

To understand:

- The potential opportunities afforded by tissue engineering and regenerative therapy
- The nature of stem cells, including somatic and adult stem cells, embryonic stem cells and induced pluripotent stem cells
- The role and range of materials and scaffolds for tissue engineering
- The role and range of molecules and their delivery
- The main challenges, safety issues and future directions

INTRODUCTION

Tissue engineering and regenerative medicine are relatively new but rapidly expanding multidisciplinary fields relevant to surgery that have the potential to transform the treatment of a wide range of human diseases. The ability of tissues to undergo spontaneous repair and regeneration is highly variable and, in most cases, limited. This has driven the development of approaches that harness the biology at the site of tissue damage to mediate regeneration through the localised delivery of cells, materials and molecules. The continuing improvement in our understanding of how tissues are formed and how they heal underpins the continuous development of novel approaches. As these technologies improve, translate and build up clinical evidence of effectiveness, the prospect of actual tissue regeneration, not just repair, will establish clinical utility. In this chapter, we explore the development of the tissue engineering paradigm (*Figure 4.1*) and the constituent processes that have been used to enhance tissue healing by considering cells, materials and molecules and the interplay between them, alongside key exemplars. We further highlight the opportunities, challenges and likely future directions for the field.

OPPORTUNITIES

The potential impact of tissue engineering and regenerative therapies is so far-reaching that practising surgeons should be aware of the resulting opportunities to improve patient management. Several conditions that could benefit from this approach are of particular relevance to surgeons because they are closely involved in assessment and treatment (*Table 4.1*). Selected examples include the repair or replacement of injured or diseased cartilage, skin, pancreatic islets, bladder, intestine, heart tissue, arteries, larynx and bronchus. A longer term goal in tissue engineering is the replacement of diseased whole organs such as the liver and kidney, although the technical challenges here are enormous.

Surgeons are integral to many of the multidisciplinary research teams currently undertaking translational research in this field and will play a vital role in the future delivery and

TABLE 4.1 Examples of tissues created by tissue engineering and conditions they may be used to treat.

Tissue	Conditions treated
Skin	Burns and skin defects after excision or trauma
Eye	Retinal and corneal disease
Cardiac muscle	Heart failure
Heart valves	Congenital and acquired valvular heart disease
Cartilage and bone	Degenerative and traumatic bone and joint disorders
Trachea and bronchus	Congenital and acquired stenosis and resection for malignancy
Bladder	Congenital bladder malformation and cystectomy
Anal/bladder sphincter	Incontinence
Pancreatic islets	Insulin-dependent diabetes
Large blood vessels	Atheromatous, aneurysmal and traumatic arterial disease
Oesophagus	Benign stenosis and resection for malignancy
Small intestine	Intestinal failure after surgical resection for Crohn's disease, cancer or ischaemia

Burrill Bernard Crohn, 1884–1983, gastroenterologist, Mount Sinai Hospital, New York, NY, USA, described regional ileitis in 1932.

evaluation of resulting treatments. In addition to its direct therapeutic application, tissue engineering also has the potential to provide *in vitro* tissues that can be used to model human disease and to test therapeutic drugs for efficacy and toxicity. However, it is important to emphasise that, while the potential benefit of cell therapy and tissue engineering is undeniable, there are many technical, regulatory and safety issues to be addressed for it to have wide clinical impact.

Summary box 4.1

Tissue engineering and regenerative therapies

These have potential to provide:

- Treatment for a wide range of diseases
- Clinical applications – underpinned by translational research, delivery strategies and clinical evidence
- Models to test therapeutic efficacy and toxicity

KEY AREAS OF UNDERPINNING SCIENCE

Advances in tissue engineering and more broadly regenerative medicine are underpinned by developments in both the physical and biological sciences, building on the classical tissue engineering paradigm (***Figure 4.1***). An improved understanding of developmental biology and the cues that direct stem cell fate have been key to advancement of the field. A better understanding of the stem cell niche has enabled scientists to propose changes to molecular and mechanical properties that could bring about modified cell behaviour. This has been realised by advances in materials science, which have been critical in the development of structures (scaffolds), onto and into which cells can be grown and where delivery vehicles can be used to localise cells and molecules to a specific site within the body. Notwithstanding the potential offered by these therapies, it should be emphasised that the whole field is still at a relatively early stage of development. Although there are examples where tissue engineering and regenerative therapies have already been introduced into clinical practice, for example the repair of damaged cartilage, most potential regenerative therapies have not yet entered routine surgical practice as there are considerable barriers to be overcome before this translational step can be achieved. We have divided the chapter into sections relating to cells, materials and molecules, while recognising the interplay and composite therapeutic solutions that ultimately arise.

CELLS

Although both fully differentiated cells (somatic cells) and stem cells are being used and developed for tissue engineering and regenerative therapy, most of the focus is on the use of stem cells, particularly somatic stem cells (SSCs) and induced pluripotent stem cells (iPSCs). The major features of the different cell types are listed in ***Table 4.2***.

Somatic cells

Fully differentiated specialised cells (somatic cells) obtained from normal tissues have been used for tissue engineering and regenerative therapy with some degree of success. For example, skin has been engineered using cultured epithelial cells grown *in vitro* and used to treat patients with burn injuries. Chondrocytes have been isolated, expanded *in vitro* and implanted into areas of deficient cartilage in a procedure called autologous chondrocyte implantation. Bladder wall has

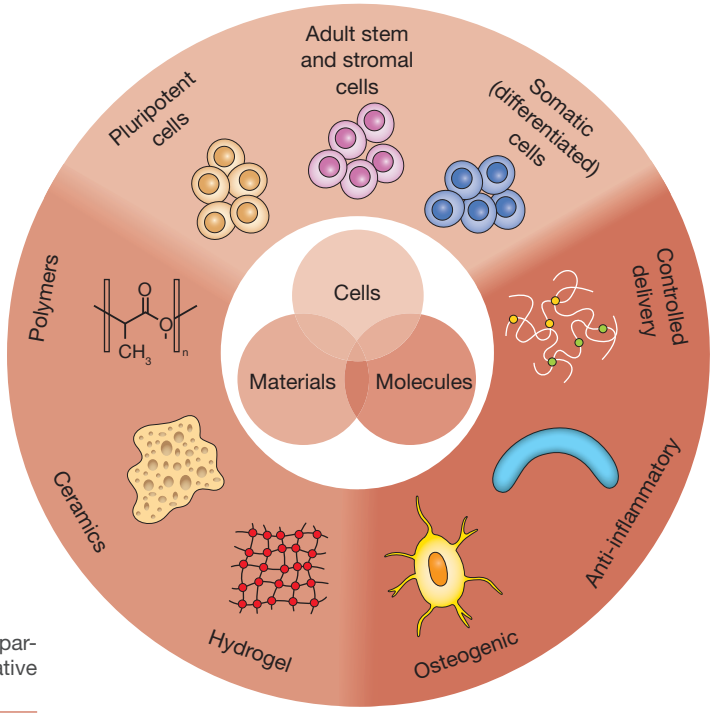

Figure 4.1 The tissue engineering paradigm in the context of regenerative therapies.

TABLE 4.2 Cells used in tissue engineering and regenerative therapy.

Cell type	Somatic cells	SSCs	hESCs	Fetal cells	iPSCs
Ease of availability	Limited	Good	Moderate	Moderate	Good
Expansion *in vitro*	Limited	Good	Excellent	Good	Excellent
Potency	No	Limited	Excellent	Limited	Excellent
Ethical concerns	No	No	Yes	Yes	Yes[a]
Risk of malignancy	None	Low	Moderate	Moderate	Moderate
Autologous	Yes	Yes	No	No	Yes
Anticipated future use	Limited	High	Limited	Limited	High

hESCs, human embryonic stem cells; iPSCs, induced pluripotent stem cells; SSCs, somatic stem cells. [a]Note that iPSCs avoid some of the ethical issues associated with hESCs.

also been engineered using a combination of smooth muscle cells and uroepithelial cells expanded *in vitro* and grown on a scaffold before reimplantation. Such tissues can be grown using cells obtained from the intended recipient by tissue biopsy (autologous cells) or using cells obtained from unrelated donors (allogeneic cells). The major advantage of the former source is that, after implantation, they are not rejected by the recipient's immune system; hence there is no requirement for immunosuppression (see ***Chapter 88***).

For other indications, the use of fully differentiated specialised cells is not practical in most situations because such cells are not readily available in sufficient numbers and they have only limited proliferative ability *in vitro*, which means that their numbers cannot be readily expanded to sufficient levels. To overcome these limitations, the major focus in the field of cell therapy has been on the use of stem cells.

Stem cells

Stem cells are undifferentiated or non-specialised cells that are able, through cell division, to renew themselves indefinitely (self-renewal). Crucially, they are also able, when provided with appropriate stimuli, to differentiate into one or more of the different types of specialised cell found in tissues and organs (potency). Because of their unique ability to undergo self-renewal when cultured *in vitro* and to be directed to differentiate into specialised cell types, they have enormous potential for use as cell-based therapies or, by way of their different characteristics, to otherwise contribute to regenerative medicine.

Stem cells can be classified in different ways, for example by using their characteristic level of potency such as pluripotent and multipotent (***Figure 4.2***) or with reference to the tissue from which they are derived, for example the early embryo

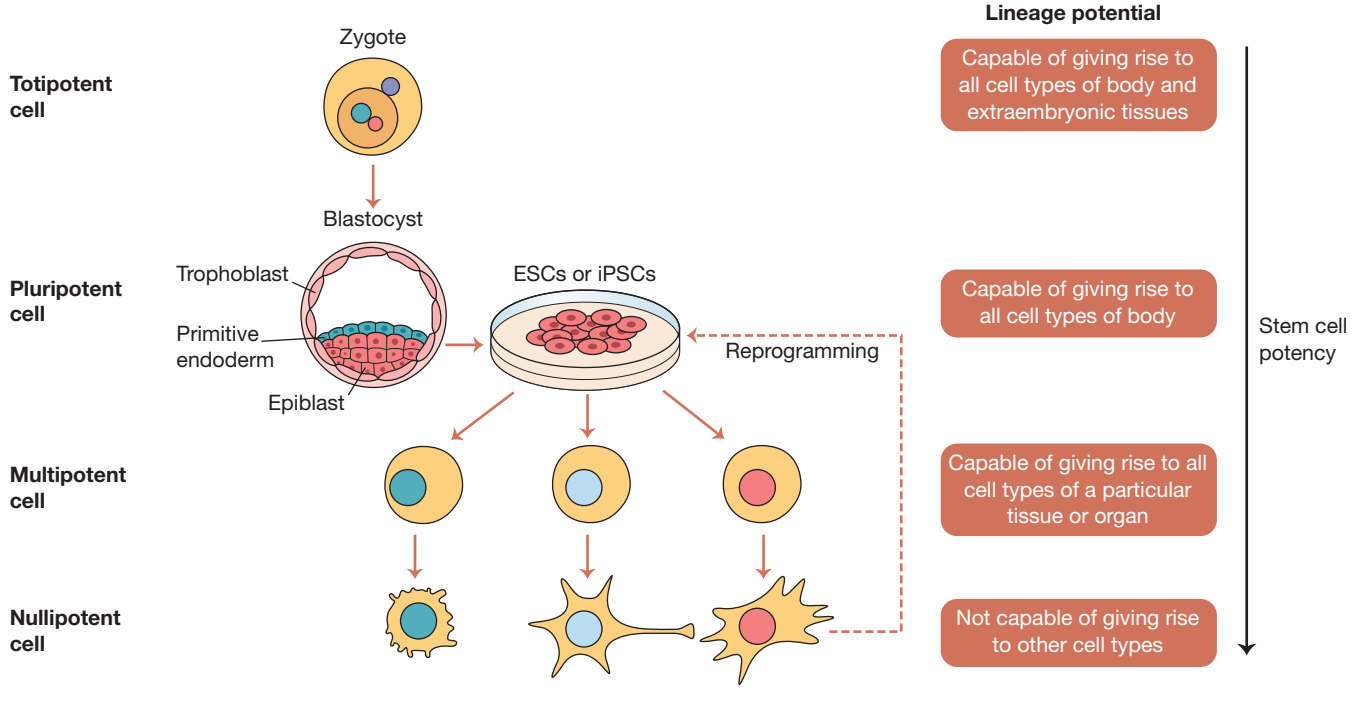

Figure 4.2 Hierarchy of cells according to potency, ranging from stem cells to specialised differentiated cells. ESC, embryonic stem cell; iPSC, induced pluripotent stem cell. (Adapted with permission from Tewary M, Shakiba N, Zandstra PW. Stem cell bioengineering: building from stem cell biology. *Nat Rev Genet 2018*; **19**: 595–614.)

(embryonic stem cells [ESCs]), later in development (SSCs) or whether they are derived by reprogramming adult specialised cells to a pluripotent state (iPSCs).

Adult tissue resident or somatic stem cells

Stem cells resident in the different tissues and organs are responsible for providing replacements for specialised cells that have reached the end of their functional lifespan either through natural attrition or because of damage and disease. In certain tissues and organs, notably the bone marrow and gut, stem cells regularly divide and differentiate into specialised cells to replace senescent or damaged cells in the blood and the gastrointestinal mucosa, respectively. Stem cells in other organs, such as the heart or central nervous system, are less able to effect repair or replacement. SSCs have the capacity to differentiate into a limited number of specialised cell types (multipotent); among the best characterised types are haematopoietic stem cells. In tissue engineering and regenerative medicine, mesenchymal stem or stromal cell (MSC) populations are widely described but have been more difficult to characterise, particularly in terms of stem cell attributes relating to clinical use.

Mesenchymal stem and stromal cells

Building on earlier observations in the 1960s[1] relating to bone marrow-derived cell populations, in the 1990s the term 'mesenchymal stem cell' (MSC) was used in relation to therapy, based on observations that some of these cells, under the right conditions, could differentiate into cell types relating to musculoskeletal, adipose and other tissues.[2] In 2005 the International Society for Cellular Therapy (now the International Society for Cell and Gene Therapy; ISCT) proposed that these cells be termed multipotent 'mesenchymal stromal cells' with the same abbreviation MSC, and that the term mesenchymal stem cell should be 'reserved for a subset of these (or other) cells that demonstrate stem cell activity by clearly stated criteria'.[3] They went on to describe minimum criteria in 2006[4]: adherence to plastic, expression of certain surface markers (CD105, CD73 and CD90) but a lack of expression of others (CD45, CD34, CD14 or CD11b, CD79α or CD19 and HLA-DR) and, finally, the ability to differentiate into osteoblasts, adipocytes and chondroblasts *in vitro*, often described by authors as trilineage differentiation.

The importance of nomenclature relates to the mechanism by which such cells might achieve a clinical effect. The earlier term mesenchymal 'stem' cell implies that cells directly contribute to repair and regeneration by differentiation, whereas using the term 'stromal' can encompass paracrine and secretory behaviour, in which cells are envisaged to work with other cells to influence the outcome of repair and regeneration (***Figure 4.3***). As a consequence, the term mesenchymal 'stem' cell has been highlighted as a cause of potential confusion, whereby patients might wrongly infer that the cell constitutes a 'stem cell therapy'.[5,6] In 2019, the ISCT gave continued support for the term mesenchymal stromal cell but recommended that it be: supplemented with the tissue source of the cell; intended unless rigorous evidence for stemness exits; associated with robust functional assays demonstrating properties.[7]

A further consideration is the manufacture and delivery of such cells. MSCs can be isolated from bone marrow (iliac crest aspiration) or from subcutaneous fat (liposuction/lipoaspiration). Cells can be delivered at the point of care, using bedside systems, or isolated *in vitro* on the basis of their adherence to plastic and subsequently further characterised. Therefore, they can be used shortly after extraction or after expansion of their numbers by *in vitro* culture. Furthermore, MSCs can be differentiated into the desired lineage *in vitro* by addition of suitable growth factors and chemicals.

The wide variety of cell type, source and manufacturing process represent important opportunities for treatment.

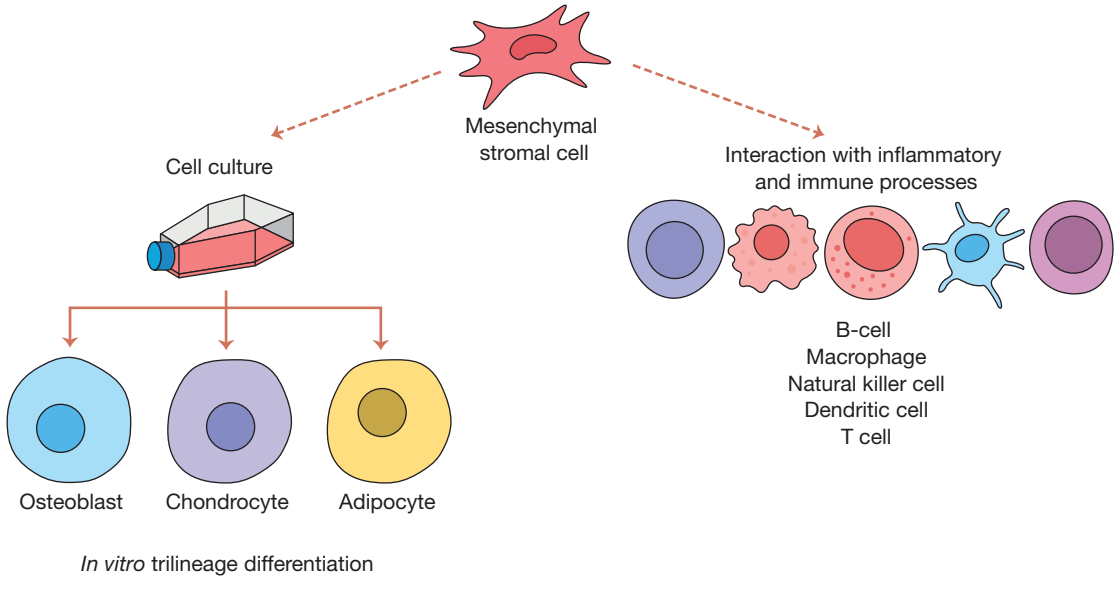

Figure 4.3 Proposed characteristics of mesenchymal stromal cells relevant to tissue engineering and regenerative medicine.

However, they also demonstrate considerable heterogeneity, the understanding of which will be greatly improved by molecular biology techniques and functional assay. In terms of agreed nomenclature, a report on consensus has described key parameters in the abbreviation DOSES: D – donor, O – origin tissue, S – separation method, E – exhibited characteristics, S – site of delivery.[8]

The relative ease of cell acquisition has meant that autologous MSCs have been used in clinical settings and they represent a great opportunity for new treatment development. Before widespread adoption, more translational research is required to understand and refine the therapeutic mechanism of action and conduct well-designed clinical trials to establish the evidence of effectiveness.

Embryonic stem cells

In the embryo, stem cells are able to give rise to all of the different cell types of the body at the gamete stage (totipotent) and nearly all cell types (pluripotent) when at the blastocyst stage (***Figure 4.1***). ESCs can be obtained from the inner cell mass of the early human blastocyst (days 4–5), using embryos that have been created through *in vitro* treatment of infertility and are surplus to those needed for reimplantation. The technique for isolating and growing human ESCs in culture was developed by James Thomson in 1998.[9]

ESCs have proliferative ability and possess key characteristics of self-renewal and pluripotency. However, their use in the clinic has major limitations, one of which is ethical. The surplus embryos used for derivation of ESCs would otherwise be discarded but, because they need to be destroyed to obtain ESCs, the approach has raised major ethical and political debate. The dominant view in many countries, including the UK, is that the potential therapeutic benefits of ESCs justify their use but there are very strict guidelines for their derivation; to date their clinical use has been limited. Cells from ESCs would be allogeneic and therefore be at risk of immunological rejection.

Induced pluripotent stem cells

The discovery in 2006 by Shinya Yamanaka,[10] building on the earlier work of John Gurdon,[11] that certain types of specialised adult cells could be reprogrammed using genetic manipulation to become embryonic-like iPSCs was a major breakthrough. Using retroviral or lentiviral transfection to introduce a combination of transcription factors (OCT3/4, SOX2 and either Kruppel-like factor and C-MYC [together designated the OSKM reprogramming factors] or NANOG and LIN28), it was shown that specialised somatic cells can be reprogrammed to become stem cells. Moreover, iPSCs proliferate *in vitro* as efficiently as ESCs and are pluripotent, thereby circumventing concerns about the use of human embryos. Importantly, the development of iPSCs also means that, at least in principle, an intended recipient of stem cell therapy can themselves provide a source of stem cells (e.g. from a skin biopsy or blood sample) that can then be directed to differentiate into the desired specialised cell type for therapy; because such cells would be autologous they would not provoke an immunological rejection response (***Figure 4.4***). Alternatively, iPSCs could be obtained from a number of volunteer donors selected on the basis of their HLA type and stored to create a national or international tissue bank of iPSCs. Lines of iPSCs could then be chosen from the bank to provide a fully or partially matched cell transplant for recipients, eliminating or reducing the need for immunosuppression to prevent immunological rejection.

One of the problems of reprogramming somatic cells to become iPSCs using retroviruses is that genomic integration of the virus may lead to activation of oncogenic genes, causing tumorigenesis. To reduce this risk, non-retroviral vectors have been used (such as adenovirus and Sendai virus vectors, which do not insert their own genes into the host cell genome) or plasmids, episomal vectors and synthetic RNA. There has also been much recent progress in identifying combinations of small molecules, growth factors and chemicals that mimic the effect of viral transfection with transcription factors and obviate the need for viral vectors altogether. The production process from sourcing cells (e.g. skin fibroblasts or peripheral blood mononuclear cells) to obtaining an adequate number of validated iPSCs may take several weeks.

In vitro differentiation of stem cells to specialised tissue cells

There is an enormous research effort aimed at better understanding the factors responsible for cell fate decisions and establishing effective and reproducible protocols that can be used to differentiate stem cells *in vitro* into the desired type of specialised cell. Typically, such protocols use culture in chemically defined media containing cocktails of small molecules that stimulate or inhibit key signalling pathways, along with cytokines, growth factors and chemicals. It is becoming increasingly clear that exposure to certain biomaterials and the physical attributes of a scaffold, including its surface characteristics, also promote stem cell differentiation along a particular lineage. Mechanical stress also influences cell fate decisions. After stem cells have been subjected to *in vitro* differentiation, it is essential that the purity of the differentiated cells and the absence of undifferentiated stem cells are confirmed to reduce the risk of tumour transmission. The cells must also be fully characterised and their function confirmed before they are used for therapeutic purposes.

Exemplars: cells as a therapy

The delivery of cells into damaged tissues has long been used to facilitate enhanced regeneration. The first modern example of successful cell therapy was bone marrow transplantation for the treatment of leukaemia, which was successfully performed in 1956. Since then, cells have been used for the regeneration

James Thomson, b. 1958, Professor and Director of Regenerative Biology, Morgridge Institute for Research, University of Wisconsin-Madison, Madison, WI, USA.
Shinya Yamanaka, b. 1962, Japanese stem cell researcher, winner of the Nobel Prize in Physiology or Medicine in 2012.
Sir John Bertrand Gurdon, b. 1933, British developmental biologist, winner of the Nobel Prize in Physiology or Medicine in 2012 with Shinya Yamanaka.

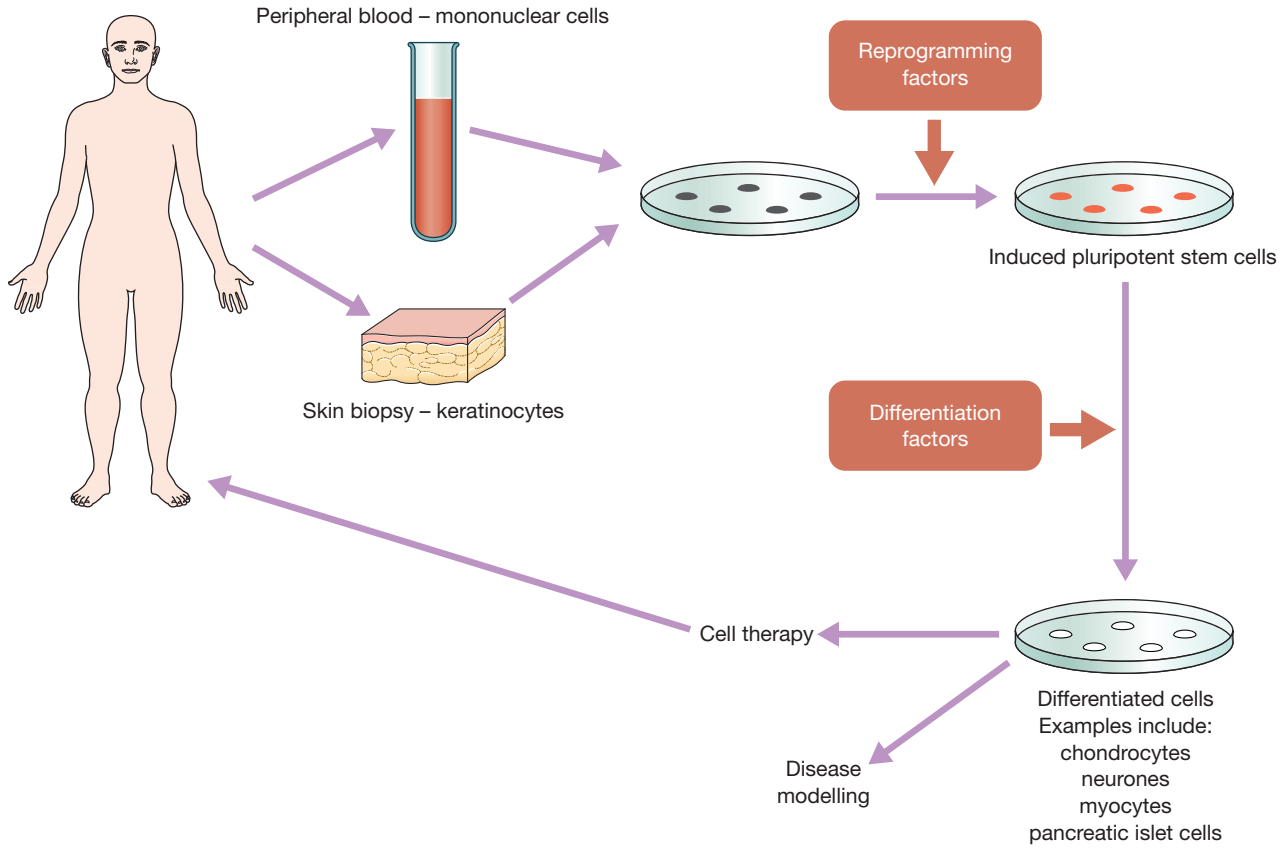

Figure 4.4 Schematic diagram showing the principles of induced pluripotent stem cell (iPSC) therapy. Mononuclear cells from peripheral blood or keratinocytes from a skin biopsy are cultured *in vitro* and then reprogrammed to become iPSCs by addition of reprogramming factors. The iPSCs are then expanded and selected differentiation factors added to promote differentiation of iPSCs into the desired specialised cell type for use as therapy.

of a range of different tissues. The following section details exemplars of the use of cells as therapeutic agents, for the regeneration of skin and cartilage (articular and auricular).

Skin

The tissue that was one of the first regenerated using human cells was skin. The pioneering work of Rheinwald and Green in this area led to the identification of optimal conditions for the growth and maintenance of keratinocytes harvested from skin in culture. The first tissue engineered skin consisted of a layer of keratinocytes grown on a collagenous membrane, which could be directly applied to a healing wound. In the original skin tissue engineering process, the keratinocytes were grown onto a membrane and were cultured in the same system as a layer of confluent but (unable to divide) fibroblasts. The fibroblasts are required in this culture because they provide a series of paracrine factors that allow for the keratinocytes to maintain their phenotype and also mature when lifted to the air–liquid interface. The fibroblasts had to be mitotically inhibited because they tend to proliferate at a greater rate than the keratinocytes and can consequently overtake the complex culture. As a consequence of the need for multiple steps and cell types, in addition to the air-lift required for stratification of the epidermal keratinocyte layer to occur, the process is relatively slow (requiring up to a month). This means that the burn wound to be resurfaced using this process needs to be closed and protected prior to application. This process has been refined over the years and has resulted in the development of a range of technologies, some of which are still on the market. Despite a reasonable level of integration, the skin itself lacks many of the features of normal human skin, for example coloration, and is not in widespread use in the clinic.

Articular cartilage regeneration

Microfracture or microdrilling (***Figure 4.5***) into subchondral bone at the base of a chondral defect creates a communication between the intra-articular and subchondral spaces, allowing blood, containing bone marrow stromal cells, to arrive at the defect. Used clinically for more than 30 years, the formation of the clot creates an environment that allows for the reformation of cartilage within the defect, which subsequently provides a level of mechanical function. Although widely used in the clinic with success, the cartilage formed is typically fibrocartilage, which is mechanically inferior to the native articular cartilage and lacks longer term durability.

James George Rheinwald, b. 1948, American research scientist.
Howard Green, 1925–2015, George Higginson Professor of Cell Biology, Harvard Medical School, Boston, MA, USA, pioneer in the science of skin regeneration.

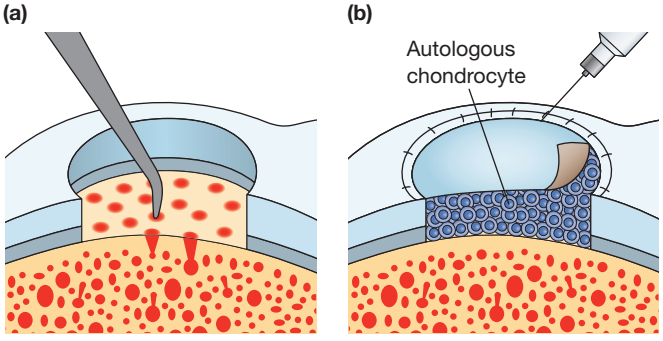

Figure 4.5 Examples of (a) endogenous cell targeting (microfracture) and (b) cells being used as a therapy (autologous chondrocyte implantation). (Adapted with permission from Makris E, Gomoll A, Malizos K *et al*. Repair and tissue engineering techniques for articular cartilage. *Nat Rev Rheumatol* 2015; **11**, 21–34.)

To overcome these limitations, research has focused on ways to create an environment in the chondral defect that is more conducive to the formation of articular or hyaline cartilage. Autologous chondrocyte implantation (ACI) is one such method (***Figure 4.5***). In the first step a small number of cells are harvested from healthy cartilage within a patient's joint, for example the knee. The cells are then expanded, to increase the cell number, in a laboratory (with good manufacturing practice [GMP] quality standards). In a second procedural step, cells are reintroduced into the defect, classically with a surgically positioned 'roof' or patch over the defect, prior to injection of the cells below. In 2017, following the evaluation of evidence, the UK's National Institute for Health and Care Excellence (NICE) recommended ACI for use in the National Health Service (NHS). The recommendation was for defined patients who have symptomatic articular cartilage defects of the knee, with a cost-effectiveness estimate thought likely to be less than £20 000 per quality-adjusted life year (QALY) gained.

MATERIALS

A large number of materials have been used in tissue engineering and regenerative medicine, either as delivery vehicles or as scaffolds; it is beyond the scope of this chapter to describe them extensively (see ***Further reading***). In simple terms, materials can be either natural or synthetic and further characterised by their mechanical properties, which will often need to take account of porosity. A further layer of complexity relates to specific molecules within and upon the material and the intended targeting or seeding with cells. In this section, we will focus on synthetic polymers, hydrogels and osteoinductive ceramics such as Bioglass®. These materials have been selected because of their broad relevance in tissue engineering and regenerative medicine.

Exemplars: materials in development

Synthetic and engineered polymers

Synthetic polymers are those made from chemical processing of natural components or from complex synthesis from precursors such as fossil fuels. There are a great many synthetic polymers that are used in the body, including polymethyl methacrylate (PMMA), which is used to make intraocular lenses and as a cement in joint replacement; Dacron™ (Terylene™), used in vascular grafts; and a range of resorbable materials, including poly(caprolactone) (PCL), poly(lactic acid) (PLA), poly(glycolic acid) (PGA) and copolymers thereof (PLGA). These resorbable polymers have been used in the body for some time as sutures and in other degradable structures. Moreover, they have become widely used in regenerative medicine for the production of scaffolds, on which cells can be seeded and grown prior to, during or after implantation into the body. Synthetic materials can be processed in a range of different ways in order to provide a structure optimised to the intended purpose. Porous monoliths can be produced by foaming processes, which can include but are not limited to infiltration and then expansion of supercritical carbon dioxide in order to create a foamed structure. The pore sizes of such structures may be tailored by modifying process conditions or material compositions (***Figure 4.6***).

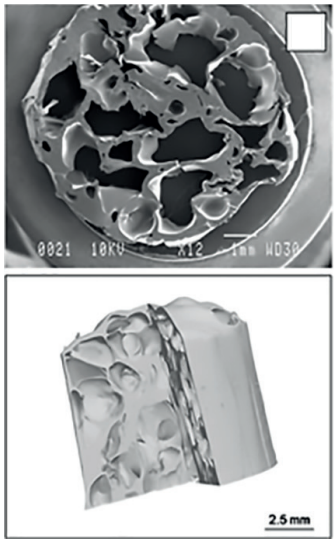

Figure 4.6 Electron micrograph of a poly(lactic acid) scaffold foamed using supercritical carbon dioxide (a) with pore structure assessed using micro-computed tomography (b). (Adapted with permission from Collins NJ, Leeke GA, Bridson RH *et al*. The influence of silica on pore diameter and distribution in PLA scaffolds produced using supercritical CO_2. *J Mater Sci: Mater Med* 2008; **19**: 1497–502.)

Such polymers can also be formed as spheres by using emulsion processing or they can be processed into a mat of fibres using a process known as electrospinning, whereby charged threads of polymer are drawn using an electric field and subsequently deposited onto a surface (***Figure 4.7***). Manufactured in this way, the materials tend to have a very high surface area and can exhibit structural features across the same length scales as the fibrous components of the extracellular matrix. As a consequence, electrospun patches have been designed for the repair of anatomical structures such as the rotator cuff and have even been used to explore how modifications in matrix geometry may result in pathology.

Bioceramics

A range of ceramics have been used for the delivery of compounds that can trigger the regeneration of mineralised tissues.

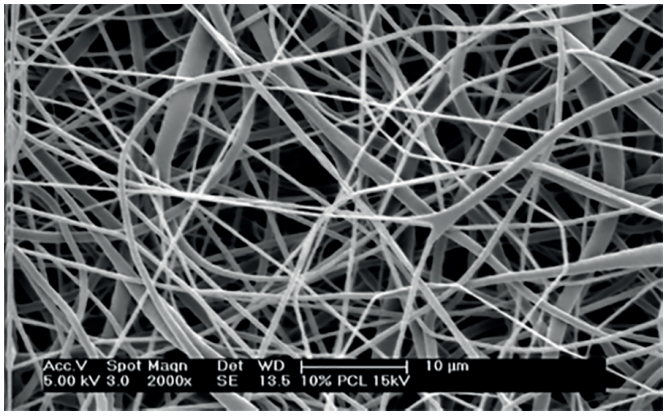

Figure 4.7 Electrospun poly(caprolactone) fibres intended for use in tendon reconstruction (courtesy of Dr Anita Ghag).

The most widely used such material is Bioglass®, a glassy material (a ceramic with no crystal structure) containing calcium, silicate and phosphate ions (Na_2O–CaO–P_2O_5–SiO_2). When placed into an aqueous environment, such as within body fluids, the surface of the ceramic material can break down, releasing the component ions into the surrounding liquid. This can result in the deposition of a bone-like mineral across the surface of the material itself, which can encourage the material to bond to surrounding hard and soft tissues. In addition, the eluted ions can trigger specific biological responses that include the recruitment and differentiation of mesenchymal cell populations. Bioglass® has been used in a range of medical products and has recently been widely applied in remineralising toothpaste formulations (as Novamin®). The incorporation of other ions, such as strontium and lithium, into the glassy matrix can drive specific biological processes, enhancing the process of bone formation. Bioglass® can be manufactured as a foamed structure and as a monolith. It is most frequently supplied in the form of granules and incorporated into polymer matrices to drive the process of bone integration.

Hydrogels

Hydrogels are an emerging and increasingly researched class of materials in regenerative medicine. They consist of hydrophilic polymers that are typically dispersed in an aqueous component to form a hydrocolloid. Interactions between the individual polymer chains can then be formed by modifying the temperature, adding ions or adding a chemical cross-linking agent. The resulting network retains water and forms a solid but highly hydrated structure (typically *c.* 99wt% water). The high water content of these materials means that nutrients, oxygen and metabolic by-products can diffuse through them. As a consequence, they have been widely used for the encapsulation of cells, typically allowing the maintenance of high levels of viability. Alginate, a seaweed-derived polysaccharide blend, for example, has been used to protect pancreatic islets from immunological attack. In recent years, this approach has been refined to allow for pancreatic islets to be shipped between medical centres (***Figure 4.8***). Other materials that form gels include chitosan, gellan, collagen and hyaluronic acid. In addition to these biologically derived materials, hydrogels may also be formed from synthetic polymers, including poly(ethylene glycol), and these can be chemically modified to provide specific biological stimuli to entrapped cells.

Additive layer manufacturing and hydrogels

As described above, hydrogels have been used for cell encapsulation. The potential to modify both the chemical and mechanical properties exhibited by these materials makes them perfect candidates for the growth of tissues outside the body, prior to eventual implantation into a defect site inside the body. As a consequence, hydrogel-based structures have provided scaffolding for many different engineered tissues, ranging from bone through to brain. However, a major issue

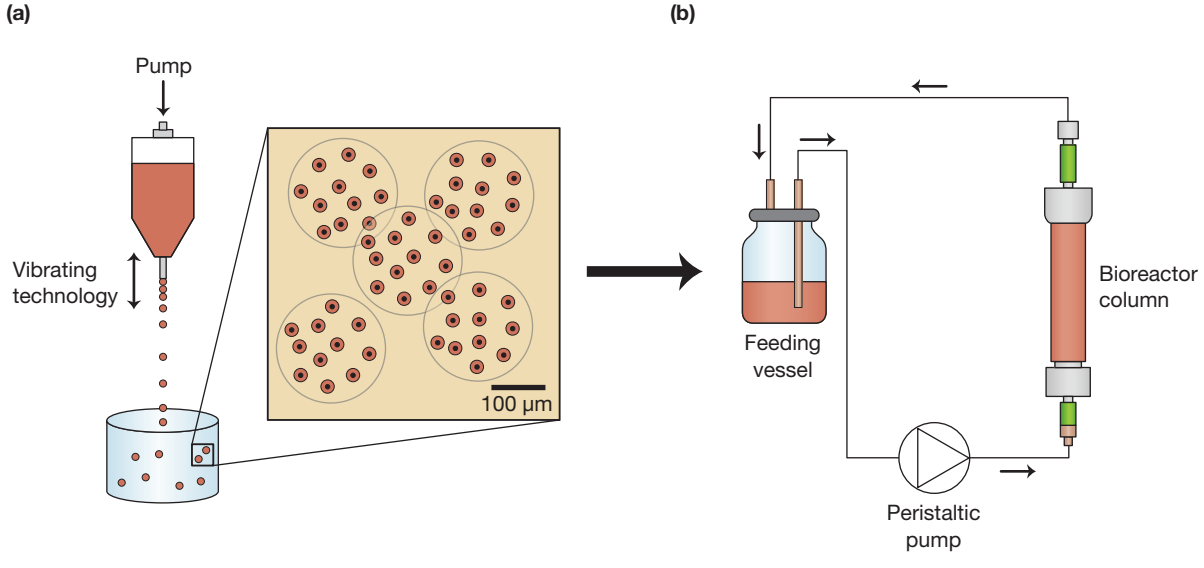

Figure 4.8 Schematic of **(a)** pancreatic β-cell encapsulation and **(b)** bioreactor cultivation. (Adapted with permission from Nikravesh N, Cox SC, Ellis MJ, Grover LM. Encapsulation and fluidization maintains the viability and glucose sensitivity of beta-cells. *ACS Biomater Sci Eng* 2017; 3(8): 1750–7.)

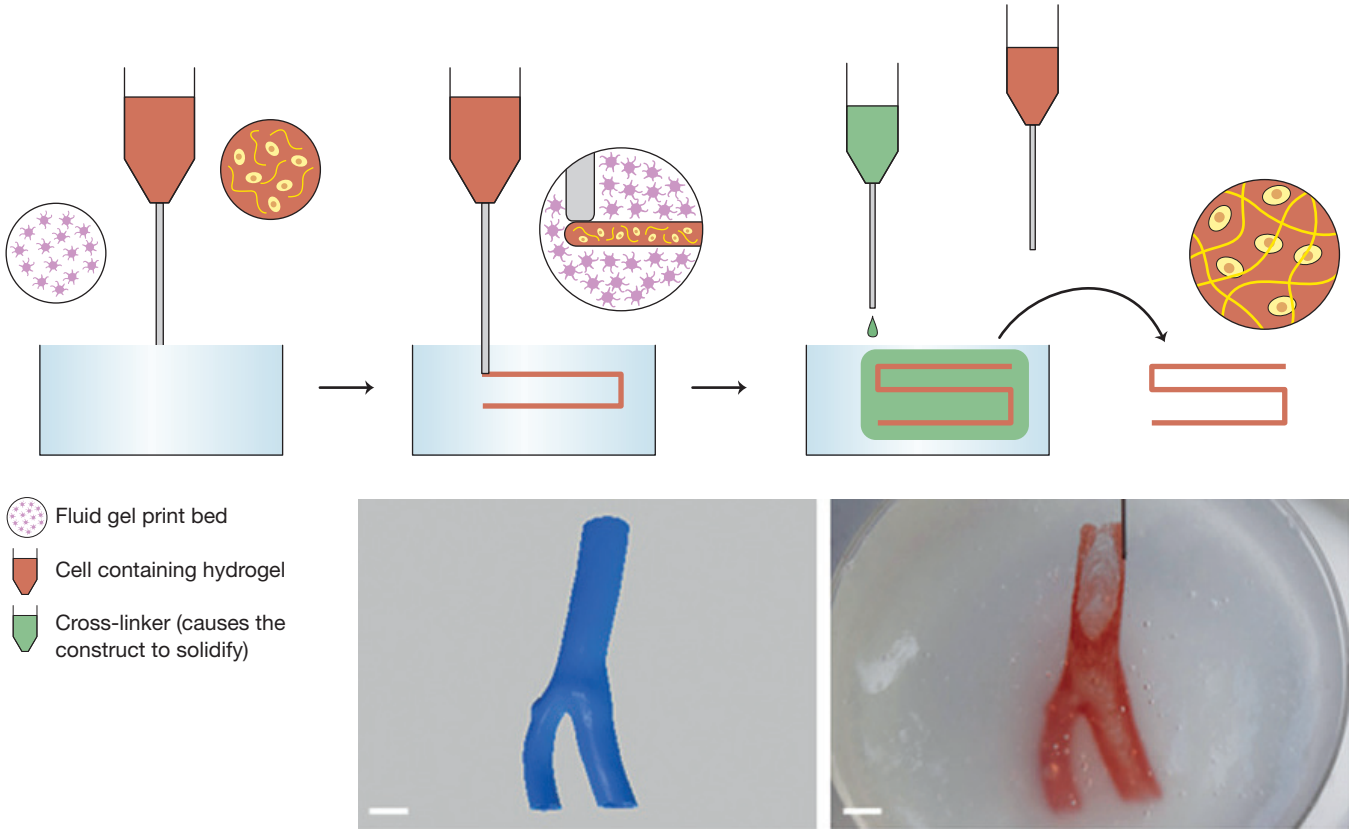

Figure 4.9 Schematic showing the formation of complex hydrogel structures using a suspended manufacturing process where a cell containing hydrogel is deposited into a supportive bed. This process allows for the production of hydrogels in complex geometries such as the carotid arteries shown here. (Adapted with permission from Senior JJ, Cooke ME, Grover LM, Smith AM. Fabrication of complex hydrogel structures using suspended layer additive manufacturing (SLAM). *Adv Funct Mater* 2019; **29**: 1904845.)

with hydrogels is that they are normally extremely weak during the gelation process, meaning that they will slump and lose their shape when cast. In order to get around this major issue, printed structures may be suspended in a secondary (often shear thinning) medium that provides both a level of support during the printing process and time for the hydrogel to develop sufficient mechanical integrity to support its own weight. Once this point has been reached, the structure can be removed from the supporting medium and washed prior to onward culturing. A wide range of substances have been used to provide support, including Pluronic® and agarose; however, the most widely used method currently is known as free-form reversible embedding of suspended hydrogels (FRESH), which uses gelatin. The FRESH method utilises a macerated gelatin-based support matrix. The particulate nature of the gelatin means that the printed hydrogel (which may or may not contain cells) can be deposited within the bed and will be supported while the mechanical properties exhibited by the as-deposited hydrogel become fully developed. The supporting matrix can then be melted (at 37°C) to leave the complex hydrogel structure. Other suspended printing methods have been developed that do not require the use of an animal-derived support medium (and that are therefore compatible with translation to the clinic) or the elevation of temperature to remove the supporting phase. One such method, suspended layer additive manufacturing (SLAM), uses a supportive matrix formed from hydrogel particulate systems that shear thin but exhibit rapid elastic recovery; this allows for the production of high-resolution prints that can be removed from the supporting bed by the application of a simple wash with water. This method enables the production of structures exhibiting complex morphology, such as the carotid artery (***Figure 4.9***), and can even be used to produce structures formed or modified from multiple material types, meaning that structures can be tailored with high resolution. With clinical relevance, SLAM has been used to manufacture osteochondral plugs containing both bony and cartilaginous regions. At present, these technologies are most likely to have an impact on disease modelling and drug candidate evaluation, but in the longer term they may emerge as key technologies facilitating tissue regeneration.

MOLECULES

Modification of the tissue environment during healing can be achieved by the delivery of molecules that are selected and able to specifically modify biological responses at the site of injury. In terms of tissue regeneration, these molecules can modulate the inflammatory environment, either by enhancement or by

inhibition of specific biological response pathways, such as angiogenesis. This approach has been used to optimise tissue regeneration across several applications, with specific examples, for the skin and the eye, described below.

Exemplars: molecules in action

Healing skin without scarring

All tissues in the body, when damaged, are subject to a conserved process by which cells are recruited to the wound to remove damaged tissue fragments and then deposit and subsequently remodel the tissue (see *Chapter 3*). In the case of the skin, the initial damage results in bleeding into the wound followed by haemostasis and, when the platelets within the clot degranulate, the secretion of proinflammatory factors. The tumour necrosis factor alpha (TNFα), interleukin (IL) 1 and IL-6, platelet-derived growth factor (PDGF), fibroblast growth factor (FGF), insulin-like growth factor 1 (IGF-1) and vascular endothelial growth factor (VEGF) recruit monocytes and endothelial cells into the wound bed to begin the wound-healing process. These cells release further cytokines, such as transforming growth factor beta 1 (TGFβ1), that mediate the formation of new extracellular matrix (normally initially collagen III in skin). Over time, the collagen III is remodelled and replaced with collagen I, which is normally deposited in a basket weave pattern. This process very efficiently closes wounds and, when not too severe, healing can be with minimal scar formation. In the case of severe damage or infection, the inflammatory response can 'overshoot', resulting in the overproduction of cytokines such as TGFβ1, which drive the recruitment of excess myofibroblasts to the wound site. This leads to the deposition and subsequent overcontraction of collagen matrix, which can result in the classic appearance of a scar.

A number of interventions have targeted this process with the aim of preventing the formation of a scar on the skin. Of note was the identification and isolation of TGFβ3, which plays a role in regulating extracellular matrix formation. This molecule was shown to regulate wound healing by blocking the TGFβ1 and -2 receptors on the cell surface and by modulating the penetration of cells in a newly forming tissue. TGFβ3 was subsequently developed as a therapeutic agent for use within the skin, with the aim of preventing scar formation. As TGFβ3 is a large, protein-based drug it was challenging to manufacture and purify and so was expensive in comparison with small-molecule drugs. Although this molecule had significant potential as an antiscarring agent in phase I and II trials, ultimately it was not adopted. This is an exemplar of a translational approach, working from a proposed mechanism through to stepwise clinical trials to test effectiveness.

Molecular delivery to prevent ocular fibrosis

Another example of the function of a tissue being significantly impaired by fibrosis is the eye. When the cornea is damaged, as long as the wound is kept clean, re-epithelialisation of the surface can occur relatively quickly with full restoration of the corneal surface within 7 days or so. However, if there is an infection on the surface of the eye there can be a potent inflammatory response that results in the rapid and disordered deposition of collagenous tissue across the ocular surface. If the surface of the eye becomes cloudy as a consequence, the patient can become blind. Recent work to create a therapeutic agent to prevent ocular scarring has focused on the localised delivery of a TGFβ1 antagonist called decorin. Decorin is a proteoglycan with a high affinity for collagen I, which it decorates the surface of in normal extracellular matrix. When free in solution, decorin also has an affinity to seven different cytokine molecules, one of which is TGFβ1, and has been shown to modulate fibrosis. A decorin-containing eye drop is able to deliver a sustained dose of decorin to the surface of the cornea following damage and subsequent infection. The drop has been shown to be effective in preclinical models of ocular fibrosis, facilitating the restoration of a transparent ocular surface with a significant reduction in markers that are usually indicative of scar formation (*Figure 4.10*).

Challenges to the delivery of molecules

Despite the opportunities afforded by molecules in tissue regeneration, major challenges remain that relate to getting the therapeutic agent to the right part of the body at the right time. Small molecules tend to diffuse rapidly through excipient (delivery) materials and so are rapidly released into the tissue environment, meaning that a strategy for sustained delivery is a necessity if repeat administration of the drug molecule is to be avoided. A number of implantable materials exist that allow for the slow localised release of molecules at the desired site of application. The majority of these are dense, degradable polymers that can be implanted locally and allowed to degrade, thus enabling sustained release of the therapeutic agent. The issues are potentially even more significant for large biomolecules such as proteins, which tend to be very sensitive to local environmental changes, with associated reductions in bioactivity. They can also have extremely high activities at very low concentrations and can be extremely expensive. In addition, excipients for these therapeutic products must be chosen very carefully so that potency is maintained and storage time maximised.

SAFETY CONCERNS

The major safety concerns of cell-based therapy and tissue engineering are listed in *Table 4.3*. One of the most serious concerns is that of tumour formation and malignant transformation. The risk of tumour formation varies according to the cell type used, the genetic modification strategy used to transform the stem cells, the site of transplantation and whether the cells are autologous or allogeneic. The direct risk of tumour formation by the transplanted cells relates specifically to ESCs and iPSCs and there appears to be little risk with SSCs. The ability of stem cells to form teratomas is one of the hallmarks

TABLE 4.3 Risks of cell-based therapy.

- Tumour formation
- Genetic and epigenetic abnormalities
- Transmission of infection
- Poor viability and loss of function
- Differentiation to undesired cell types
- Rejection (allogeneic cells)
- Side effects of immunosuppression (allogeneic cells)

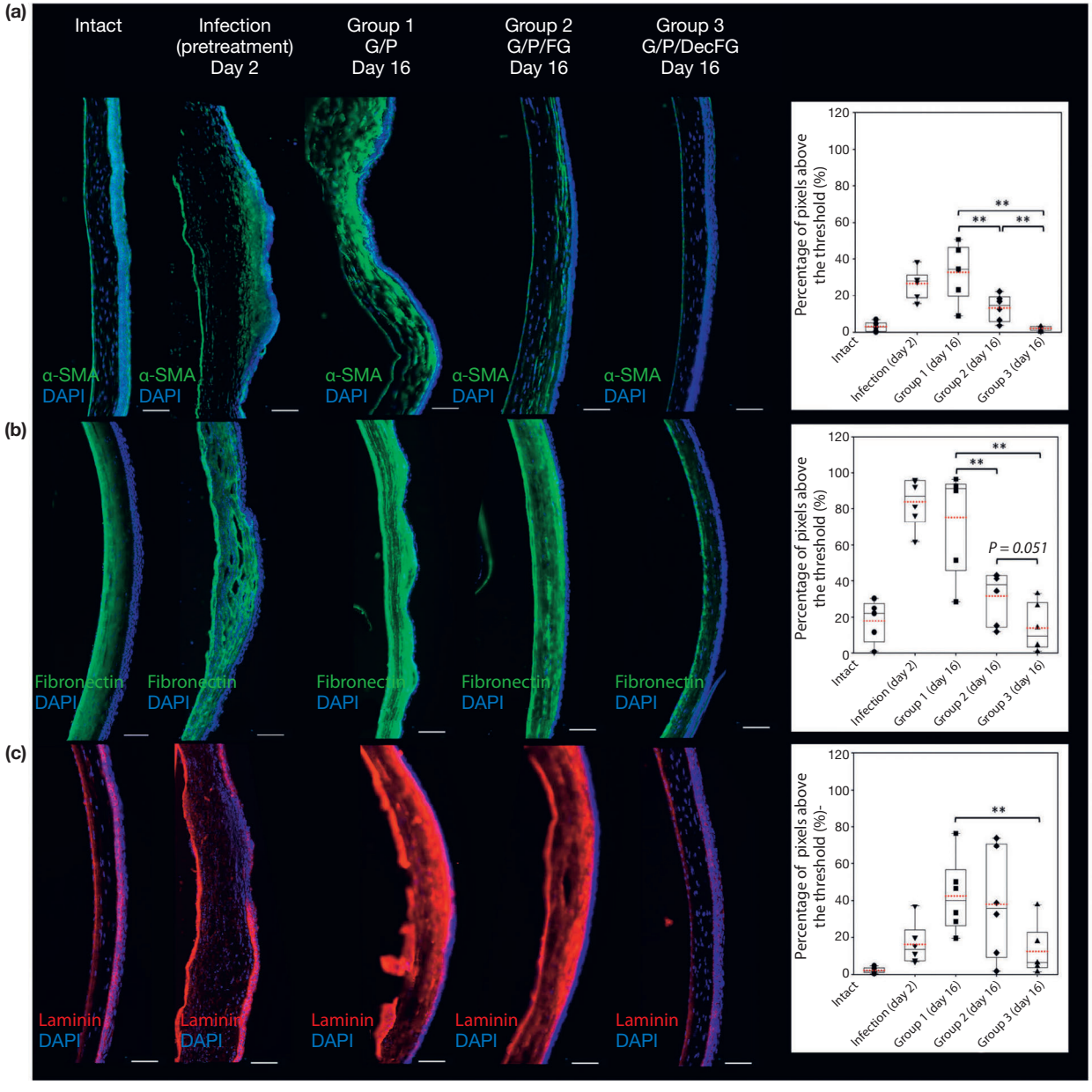

Figure 4.10 Stained sections through a mouse cornea before (left) and after (second column) injury and subsequent treatment with standard of care (gentamicin [G] and prednisolone [P]; group 1), standard of care plus a novel fluid gel carrier (FG) (group 2) and standard of care plus the carrier and decorin (Dec) (group 3). Sections are stained for markers of scarring: **(a)** α-smooth muscle actin (α-SMA), **(b)** fibronectin and **(c)** laminin. Importantly, the use of the slow-release decorin resulted in rapid restoration of the corneal structure with a significant reduction in scar markers. DAPI, 4′,6-diamidino-2-phenylindole (fluorescent stain that binds strongly to adenine-/thymine-rich regions in DNA). (Adapted with permission from Hill LJ, Moakes RJA, Vareechon C *et al*. Sustained release of decorin to the surface of the eye enables scarless corneal regeneration. *npj Regen Med* 2018; 3: 23.)

of pluripotency, and the risk of this happening following stem cell therapy may be reduced by ensuring that only cells that have been fully differentiated *in vitro*, and not those that are still pluripotent, are used for therapy. The risk of malignancy may also be reduced by the choice of *in vitro* strategy used to differentiate stem cells prior to use: the use of viral vectors that do not integrate into the genome or of non-viral approaches to differentiation reduces the risk of malignant transformation. There is also interest in developing techniques for directly reprogramming somatic cells to adopt the function of a different cell type without having to make them first revert back to the pluripotent state – so-called transdifferentiation.

Another major concern is that of transmitting infection. It is essential that if allogeneic stem cells are used they are screened to exclude infection and that cells and engineered tissues are prepared according to GMP guidelines to avoid bacterial infection during *in vitro* culture prior to use. Moreover, if allogeneic cells are used for tissue engineering and regenerative therapy, they may be susceptible to graft rejection and immunosuppressive therapy may be necessary.

FUTURE DIRECTIONS

Tissue engineering and regenerative strategies hold out great hope for effectively repairing or replacing tissues in a wide number of human diseases. The field is moving rapidly, underpinned by new developments in the relevant science in stem cells, materials and molecules. New emerging areas of technology include therapeutic signalling by way of extracellular vesicles (EVs) and gene editing of cells using CRISPR-Cas9, and gene therapies. All will require the use of a translational approach, whereby the hypothesised mechanism is developed and translated to the clinic, building up robust clinical evidence of efficacy, by way of well-designed and well-conducted clinical trials before widespread adoption.

It is likely that patient stratification will further refine therapy options. The ability to phenotype, genotype and profile patients at a molecular level will allow more detailed characterisation of patient subgroups and staging of disease. In addition to clinical studies and evidence, the rapid pace of therapy development will need to be accompanied by the development of new regulatory frameworks, for example in point-of-care manufacturing.

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CRISPR-Cas9, clustered regularly interspaced short palindromic repeats and associated protein 9. Resulted in the Nobel Prize in Chemistry being awarded to Emmanuelle Charpentier and Jennifer A. Doudna in 2020.